IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL SIGNAL PROCESSING AND DIGITAL FILTERS

Wednesday, 29 April 2026, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question ONE and any TWO other questions.

Question One is worth 40% of the marks and other questions are worth 30%.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Bhandari
Second Marker(s) : T. Stathaki

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Department of Electronic and Electrical Engineering

Examinations 2026

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

May, 2026 • Exam Duration: Three Hours

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Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the total marks
while each of the other questions is worth 30%.

If necessary, make reasonable assumptions for your solutions and clearly explain them in your answers.

A formula sheet is attached in the next two pages.

Markers: Ayush Bhandari and Tania Stathaki.

FORMULA SHEET AVAILABLE IN EXAM

The following formulae will be available in the exam:

Where a question requires a numerical answer, it must be given as a fully evaluated decimal number and not as an unevaluated arithmetic expression.

Notation

- All signals and filter coefficients are real-valued unless explicitly noted otherwise.
- Unless otherwise specified, upper and lower case letters are used for sequences and their z-transforms respectively. The signal at a block diagram node V is $v[n]$ and its z-transform is $V(z)$.
- $x[n] = [a, b, c, d, e, f]$ means that $x[0] = a, \ldots x[5] = f$ and that $x[n] = 0$ outside this range.
- $\Re(z)$, $\Im(z)$, z^*, $|z|$ and $\angle z$ denote respectively the real part, imaginary part, complex conjugate, magnitude and argument of a complex number z.
- The expected value of x is denoted $E\{x\}$.
- In block diagrams: solid arrows denote the direction of signal flow; an open triangle denotes a gain element with the gain indicated adjacently; a "+" in a circle denotes an adder/subtractor whose inputs may be labelled "+" or "−" according to their sign; the sample rate, f, of a signal in Hz may be indicated in the form "@f".

Abbreviations

BIBO	Bounded Input, Bounded Output	IIR	Infinite Impulse Response
CTFT	Continuous-Time Fourier Transform	LTI	Linear Time-Invariant
DCT	Discrete Cosine Transform	MDCT	Modified Discrete Cosine Transform
DFT	Discrete Fourier Transform	PSD	Power Spectral Density
DTFT	Discrete-Time Fourier Transform	SNR	Signal-to-Noise Ratio
FIR	Finite Impulse Response		

Standard Sequences

- $\delta[n] = 1$ for $n = 0$ and 0 otherwise.
- $\delta_{\text{condition}}[n] = 1$ whenever "condition" is true and 0 otherwise.
- $u[n] = 1$ for $n \geq 0$ and 0 otherwise.

Geometric Progression

- $\sum_{n=0}^{r} \alpha^n z^{-n} = \frac{1-\alpha^{r+1} z^{-r-1}}{1-\alpha z^{-1}}$ provided that $\alpha z^{-1} \neq 1$.
- $\sum_{n=0}^{\infty} \alpha^n z^{-n} = \frac{1}{1-\alpha z^{-1}}$ provided that $\left|\alpha z^{-1}\right| < 1$.

Forward and Inverse Transforms

z:	$X(z) = \sum_{-\infty}^{\infty} x[n] z^{-n}$	$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$
CTFT:	$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega$
DTFT:	$X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\omega n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$
DFT:	$X[k] = \sum_{0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$	$x[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$
DCT:	$X[k] = \sum_{n=0}^{N-1} x[n] \cos \frac{2\pi(2n+1)k}{4N}$	$x[n] = \frac{X[0]}{N} + \frac{2}{N} \sum_{n=1}^{N-1} X[k] \cos \frac{2\pi(2n+1)k}{4N}$
MDCT:	$X[k] = \sum_{n=0}^{2N-1} x[n] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$	$y[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$

Convolution

DTFT:	$v[n] = x[n] * y[n] \triangleq \sum_{r=-\infty}^{\infty} x[r] y[n-r]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = X\left(e^{j\omega}\right) Y\left(e^{j\omega}\right)$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = \frac{1}{2\pi} X\left(e^{j\omega}\right) \circledast Y\left(e^{j\omega}\right) \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} X\left(e^{j\theta}\right) Y\left(e^{j(\omega-\theta)}\right) d\theta$
DFT:	$v[n] = x[n] \circledast_N y[n] \triangleq \sum_{r=0}^{N-1} x[r] y[(n-r)_{\bmod N}]$	$\Leftrightarrow$	$V[k] = X[k] Y[k]$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V[k] = \frac{1}{N} X[k] \circledast_N Y[k] \triangleq \frac{1}{N} \sum_{r=0}^{N-1} X[r] Y[(k-r)_{\bmod N}]$

Group Delay

The group delay of a filter, $H(z)$, is $\tau_H(e^{j\omega}) = -\frac{d\angle H(e^{j\omega})}{d\omega} = \Re\left(\frac{-z}{H(z)} \frac{dH(z)}{dz}\right)\Big|_{z=e^{j\omega}} = \Re\left(\frac{\mathscr{F}(nh[n])}{\mathscr{F}(h[n])}\right)$ where $\mathscr{F}(\,)$ denotes the DTFT.

Order Estimation for FIR Filters

Three increasingly sophisticated formulae for estimating the minimum order of an FIR filter with unity gain passbands:

1. $M \approx \frac{a}{3.5\Delta\omega}$
2. $M \approx \frac{a-8}{2.2\Delta\omega}$
3. $M \approx \frac{a-1.2-20\log_{10} b}{4.6\Delta\omega}$

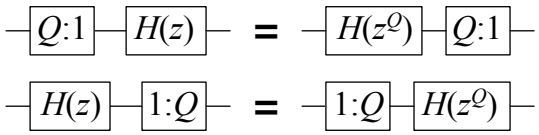
where a =stop band attenuation in dB, b = peak-to-peak passband ripple in dB and $\Delta\omega$ = width of smallest transition band in radians per sample.

z-plane Transformations

A lowpass filter, $H(z)$, with cutoff frequency ω_0 may be transformed into the filter $H(\hat{z})$ as follows:

Target $H(\hat{z})$	Substitute	Parameters
Lowpass $\hat{\omega} < \hat{\omega}_1$	$z^{-1} = \frac{\hat{z}^{-1}-\lambda}{1-\lambda\hat{z}^{-1}}$	$\lambda = \frac{\sin\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}{\sin\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}$
Highpass $\hat{\omega} > \hat{\omega}_1$	$z^{-1} = -\frac{\hat{z}^{-1}+\lambda}{1+\lambda\hat{z}^{-1}}$	$\lambda = \frac{\cos\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}$
Bandpass $\hat{\omega}_1 < \hat{\omega} < \hat{\omega}_2$	$z^{-1} = -\frac{(\rho-1)-2\lambda\rho\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\rho\hat{z}^{-1}+(\rho-1)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}, \rho = \cot\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$
Bandstop $\hat{\omega}_1 \not< \hat{\omega} \not< \hat{\omega}_2$	$z^{-1} = \frac{(1-\rho)-2\lambda\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\hat{z}^{-1}+(1-\rho)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}, \rho = \tan\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$

Noble Identities

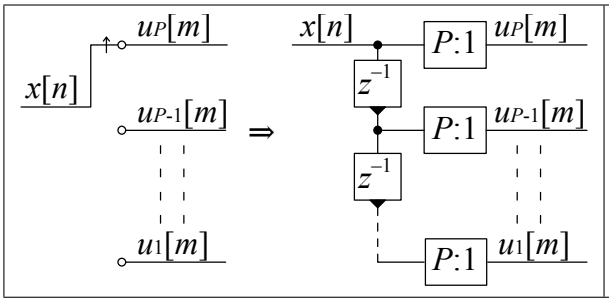
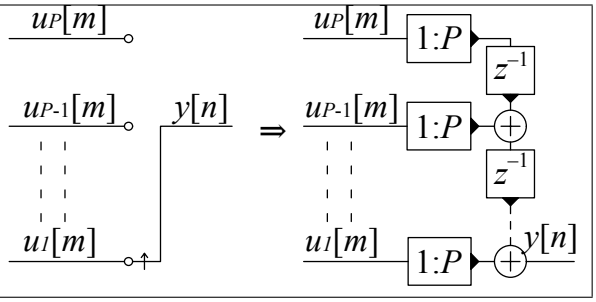


Multirate Spectra

Upsample: $v[n] \xrightarrow{1:Q} x[r] \quad \Rightarrow \quad x[r] = \begin{cases} v\left[\frac{r}{Q}\right] & \text{if } Q \mid r \\ 0 & \text{if } Q \nmid r \end{cases} \quad \Rightarrow \quad X(z) = V(z^Q)$

Downsample: $v[n] \xrightarrow{Q:1} y[m] \quad \Rightarrow \quad y[m] = v[Qm] \quad \Rightarrow \quad Y(z) = \frac{1}{Q}\sum_{k=0}^{Q-1} V\left(e^{\frac{-j2\pi k}{Q}} z^{\frac{1}{Q}}\right)$

Multirate Commutators

Input Commutator	Output Commutator
	

Question 1 [40 Marks]

Q1-1 The function $f(t) = \text{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$ defines an ideal bandlimited signal.

What is the Nyquist rate for this signal? **[2 Marks]**

Q1-2 Let

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + b_3 z^{-3} + b_4 z^{-4}}{1 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + a_4 z^{-4}}$$

be a causal LTI filter. Define a second filter $H_R(z) = H(-z)$.

Let $p[n]$ denote the impulse response of $H(z)$ and $q[n]$ denote the impulse response of $H_R(z)$, i.e., $p[n]$ and $q[n]$ are the outputs when the input is $x[n] = \delta[n]$.

(a) Write $H_R(z)$ explicitly in terms of $\{a_k\}$ and $\{b_k\}$. **[3 Marks]**

(b) Show that $q[n] = (-1)^n p[n]$. **[2 Marks]**

(c) Deduce that all even-indexed samples of the impulse response are unchanged and all odd-indexed samples change sign, i.e.,

$$q[2m] = p[2m], \qquad q[2m+1] = -p[2m+1].$$

[2 Marks]

Q1-3 Bilinear Mapping. Consider the analog filter

$$H(s) = \frac{1}{as^2 + bs + c}.$$

The filter is converted into a digital filter using the bilinear transform

$$s = \frac{2}{T}\frac{1 - z^{-1}}{1 + z^{-1}}.$$

Assume that $T = 1$ and that the resulting digital transfer function $H(z)$ has denominator

$$\alpha + \beta z^{-1} + \gamma z^{-2}.$$

i) Express α, β, γ in terms of a, b, c. **[2 Marks]**

ii) Suppose that $\gamma = 0$ and that the analog filter has unity DC gain i.e. $H(0) = 1$. Determine c and b in terms of a, and hence express $H(s)$ as a function of a. **[3 Marks]**

Q1-4 A certain system is defined by its impulse response,

$$x[n] \longrightarrow \boxed{\boxed{h_\lambda[n] = \lambda^{\lfloor \frac{n-2}{4} \rfloor} u[n-2]}} \longrightarrow y[n], \qquad 0 < \lambda < 1.$$

where $u[n]$ is the unit-step sequence and $\lfloor x \rfloor$ denotes the largest integer less than or equal to x. For example, $\lfloor 2.7 \rfloor = 2$ and $\lfloor 3 \rfloor = 3$.

(a) Compute the z-transform $H_\lambda(z)$ and state the ROC. **[4 Marks]**

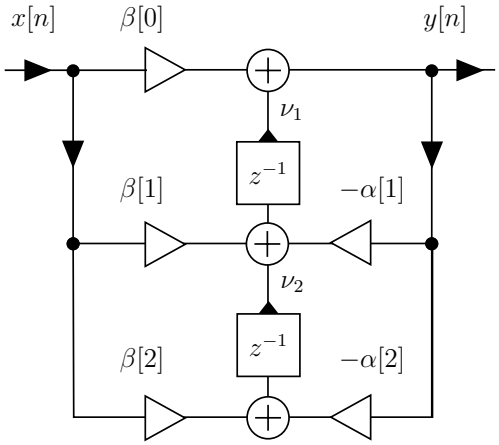
(b) Show that when $\lambda = 0$,

$$H_0(z) = z^{-2}(1 + z^{-1} + z^{-2} + z^{-3}).$$

[2 Marks]

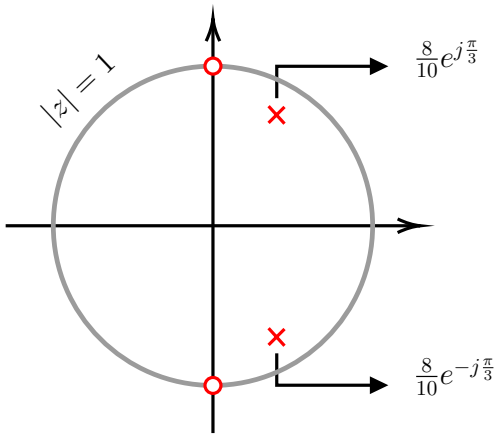
(c) Compute the impulse response $h_0[n]$ and its energy $E = \sum_{n=-\infty}^{\infty} |h_0[n]|^2$. **[2 Marks]**

Q1-5 Consider the 2nd-order biquad shown in the figure, with coefficients $\{\beta[0], \beta[1], \beta[2]\}$ and feedback taps $\{-\alpha[1], -\alpha[2]\}$.



i) Write the transfer function $H(z) = \frac{Y(z)}{X(z)}$ in terms of $\alpha[1], \alpha[2], \beta[0], \beta[1], \beta[2]$. **[2 Marks]**

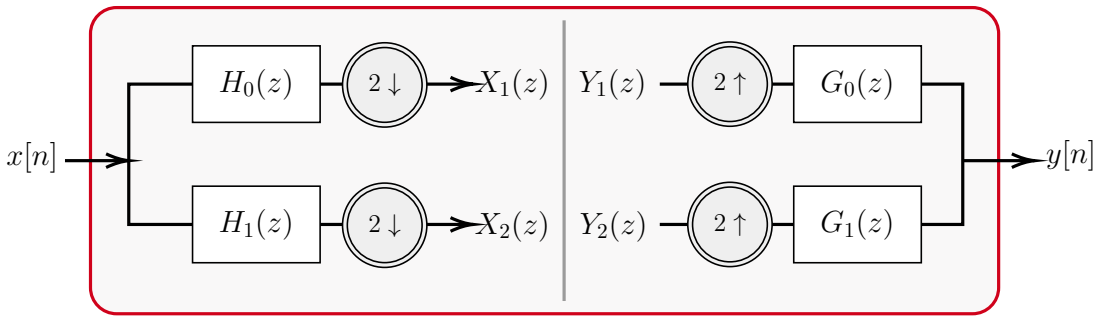
ii) Choose specific numerical values of $\alpha[1], \alpha[2], \beta[0], \beta[1], \beta[2]$ such that the filter has the following pole-zero plot:



Write the filter coefficients $\{\alpha[k], \beta[k]\}$. **[2 Marks]**

iii) Is the filter BIBO stable? Justify. **[2 Marks]**

Q1-6 Consider the two–channel analysis/synthesis system shown in the figure.



The downsampling and upsampling factors are 2. The analysis low–pass filter is

$$H_0(z) = 1 + 2z^{-1} + z^{-2}.$$

The analysis high–pass filter is $H_1(z)$ and the synthesis filters are $G_0(z)$ and $G_1(z)$.

i) Derive the overall input-output relation and show that it can be written as

$$Y(z) = \frac{1}{2}\Big[T(z)X(z) + A(z)X(-z)\Big].$$

Identify the functions $T(z)$ and $A(z)$. **[4 Marks]**

ii) State the conditions for perfect reconstruction. **[2 Marks]**

iii) Design $H_1(z)$ such that the analysis filters form a quadrature mirror pair, i.e.

$$H_1(z) = H_0(-z).$$

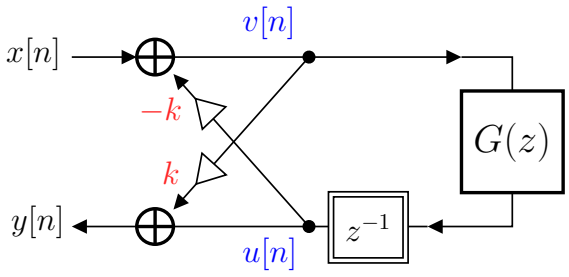
Compute $H_1(z)$. **[2 Marks]**

iv) Choose synthesis filters $G_0(z)$ and $G_1(z)$ such that aliasing is cancelled. **[2 Marks]**

v) Determine the overall system response $Y(z)$ and verify that reconstruction is achieved up to an LTI distortion proportional to $4z^{-1}(1 + z^{-2})$. **[2 Marks]**

Question 2 [30 Marks]

Q2-1 Consider the discrete-time system given by the filter structure:



The intermediate signals are defined by $v[n]$ and $u[n]$. The output is given by $y[n]$.

Assume $G(z)$ is an N-th order stable allpass filter of the form:

$$G(z) = \frac{z^{-N}A(z^{-1})}{A(z)}, \qquad A(z) = \sum_{n=0}^{N} a[n]\, z^{-n}, \quad a[0] = 1.$$

(a) Provide the time-domain representation of sequences $v[n]$ and $u[n]$. **[2 Marks]**

.

(b) Derive $H(z) = \frac{Y(z)}{X(z)}$.
Show that it can be written as

$$H(z) = \frac{k + z^{-1}G(z)}{1 + kz^{-1}G(z)}.$$

[2 Marks]

(c) For some polynomial $D(z) = \sum_{n=0}^{N+1} d[n]z^{-n}$, show that $H(z)$ can be expressed in the allpass form:

$$H(z) = z^{-(N+1)}\frac{D(z^{-1})}{D(z)}$$

Write down $D(z)$ explicitly in terms of $A(z)$, $A(z^{-1})$, and k. **[4 Marks]**

(d) By comparing coefficients, derive $d[n]$ in terms of $a[n]$ and k, for $n = 0, 1, \ldots, N+1$. **[4 Marks]**

(e) Suppose you are instead given $\{d[n]\}_{n=0}^{N+1}$. Derive how to recover k and then $\{a[n]\}_{n=0}^{N}$ from $\{d[n]\}$, assuming $|k| \neq 1$. **[4 Marks]**

(f) Assuming $G(z)$ is stable, state a necessary and sufficient condition on k for $H(z)$ to be stable. **[2 Marks]**

Q2-2 The type-II Discrete Cosine Transform (DCT) of a length-N sequence $x[n]$ is defined as

$$X_C[k] = \sum_{n=0}^{N-1} x[n]\cos\left(\frac{2\pi(2n+1)k}{4N}\right), \qquad k = 0, 1, \ldots, N-1.$$

Let

$$X(z) = \sum_{n=0}^{N-1} x[n]z^{-n}$$

denote the z-transform of the sequence $x[n]$. Show that the DCT coefficients can be written as

$$X_C[k] = \Re\left\{ e^{-j\frac{\pi k}{2N}} X\left(e^{j\frac{\pi k}{N}}\right)\right\}.$$

[6 Marks]

Q2-3 Let $h[n]$ be the impulse response of an FIR filter with frequency response

$$H(e^{j\omega}) = \sum_n h[n] e^{-j\omega n}.$$

The group delay is defined as $\tau_g(\omega) = -\dfrac{d}{d\omega} \arg H(e^{j\omega})$.

(a) Show that

$$\frac{d}{d\omega} H(e^{j\omega}) = -j \sum_n n h[n] e^{-j\omega n}.$$

[3 Marks]

(b) Using the identity

$$\tau_g(\omega) = \frac{\Re\{H^*(e^{j\omega})\, j\, \frac{d}{d\omega} H(e^{j\omega})\}}{|H(e^{j\omega})|^2},$$

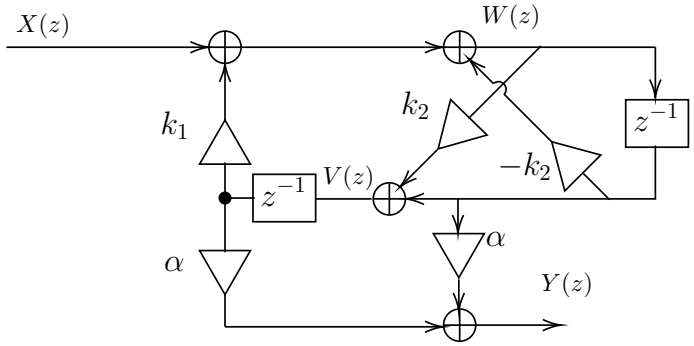
show that for a linear-phase FIR filter

$$\tau_g = \frac{\sum_n n|h[n]|^2}{\sum_n |h[n]|^2}.$$

[3 Marks]

Question 3 [30 Marks]

Q3-1 Let $H(z) = \frac{Y(z)}{X(z)}$ be a system represented in the diagram below



Suppose that $k_1 k_2 = 1$. Find the zeros of the filter $H(z)$. **[8 Marks]**

Q3-2 A filter, with input $x[n]$ and output $y[n]$, is defined by the difference equation

$$y[n] + a_1 y[n-1] + a_2 y[n-2] = a_2 x[n] + a_1 x[n-1] + x[n-2],$$

where $a_1, a_2 \in \mathbb{R}$. Assume the system is stable.

(a) Determine the system function $H(z)$ of the filter. **[3 Marks]**

(b) Show that the filter is an all-pass filter, i.e. $|H(e^{j\omega})| = 1$ for all ω. **[2 Marks]**

(c) Determine the locations of the poles and zeros of $H(z)$ in terms of a_1 and a_2, and briefly describe the relationship between them. **[3 Marks]**

(d) Let $\tau_g(\omega)$ denote the group delay of the filter.
Determine conditions on a_1 and a_2 such that

$$\tau_g(0) = \tau_g\left(\frac{\pi}{4}\right).$$

[4 Marks]

(e) Under the condition obtained in part (d), sketch qualitatively the pole–zero pattern of the filter and explain how the symmetry of this pattern leads to the observed behaviour of the group delay. **[2 Marks]**

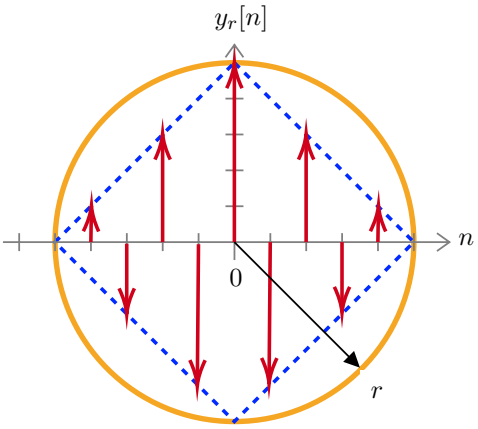
Q3-3 Let the system $G(z)$ be defined as

$$G(z) = \frac{5 - z^{-1}}{3 - (1+\alpha)z^{-1} + z^{-2}}, \qquad \alpha \in \mathbb{R}.$$

Find all α such that $G(z)$ has two *distinct real* poles inside the unit circle. **[8 Marks]**

Question 4 [30 Marks]

Q4-1 Let $r > 0$ be a real-valued number.



Calculate the energy of the signal $y_r[n]$. **[6 Marks]**

Show that when r is an integer, the energy reduces to:

$$E(r) = \frac{2r^3 + r}{3}.$$

[2 Marks]

Hint: Use the formulas to simplify your calculations:
$\sum_{k=0}^{K-1} k = \frac{K(K-1)}{2}$ and $\sum_{k=0}^{K-1} k^2 = \frac{K(K-1)(2K-1)}{6}$.

Q4-2 Let $S(z)$ be the z-transform of a sequence $s[k]$, and suppose that

$$(1 - z^{-1})S(z) = 1 + z^{-1}.$$

i) Determine the sequence $s[k]$. **[3 Marks]**

ii) Evaluate the frequency response $S(e^{j\omega})$. **[2 Marks]**

iii) Let $\overline{h}[n] = h[-n]$ and define

$$h_E[n] = \frac{1}{2}\big(h[n] + \overline{h}[n]\big), \qquad h_O[n] = \frac{1}{2}\big(h[n] - \overline{h}[n]\big).$$

Show that

$$\Re(H(e^{j\omega})) = H_E(e^{j\omega}), \qquad \Im(H(e^{j\omega})) = -jH_O(e^{j\omega}).$$

[2 Marks]

iv) Suppose that $h[n]$ is causal. Show that

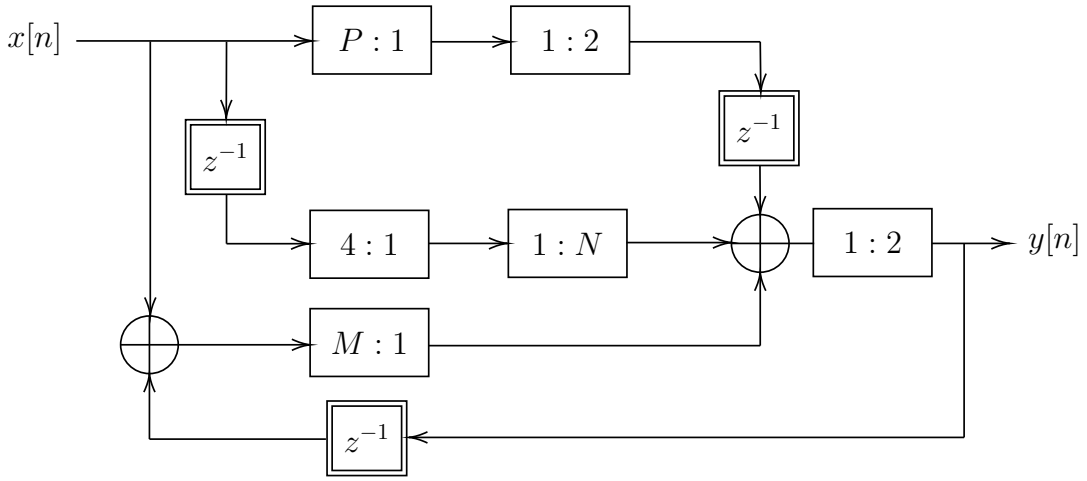
$$h_O[n] = h_E[n]\, s[n].$$

Hence show that

$$\Im(H(e^{j\omega})) = -\,\Re(H(e^{j\omega})) \odot \cot\left(\frac{\omega}{2}\right),$$

where $\odot$ denotes circular convolution. **[3 Marks]**

Q4-3 Let $x[n]$ be the input and $y[n]$ the output of a multirate system as defined in the diagram below



i) Calculate M, N, P such that all computations above are valid. **[2 Marks]**

ii) Derive the difference equation of the system. **[4 Marks]**

iii) Design a generalised linear phase filter $H(z)$ that cancels out a harmonic of frequency $\omega_0 = \frac{\pi}{2}$. Additionally, ensure that the gain at $\omega_1 = \frac{\pi}{4}$ is 1 and the gain at $\omega_2 = \frac{3\pi}{4}$ is 5. **[6 Marks]**